

Kaden Ren

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

B.S. Electrical Engineering, Robotics & Autonomous Systems

May 2028

- **Stamps President's Scholar:** Full-ride merit scholarship; selected among 25 of 37,300 out-of-state applicants.

EXPERIENCE

Undergraduate Researcher

Sep. 2026 – Present

Lunar Lab, Georgia Institute of Technology | Prof. Lu Gan

Atlanta, GA

- Research focus: robotic perception and autonomous navigation in unstructured environments.

Machine Learning Engineering Intern

Sep. 2024 – Aug. 2025

Hayden AI

San Francisco, CA

- Developed machine learning models and computer vision solutions for large-scale transit data and urban safety.

Research Intern

Oct. 2022 – Dec. 2025

Quantum Structures & Wheeldon Laboratories, UC Riverside

Riverside, CA

- Engineered semiconductor thin films using molecular beam epitaxy (MBE) for bandgap and band structure studies.
- Co-authored *Scientific Reports* research on ultra-wide-bandgap zinc gallium oxide thin films grown by plasma-assisted MBE across Zn concentrations of 0.0%–48.0%.

Research Intern

June 2025 – Nov. 2025

Garcia Center for Polymers at Engineered Interfaces, Stony Brook University

Stony Brook, NY

- Fabricated PS/PLA blend thin films by spin-casting to study morphology; presented at MRS 2025, ACS 2026, and APS Global Physics Summit 2026.

Research Intern

July 2024 – Aug. 2024

UC COSMOS Research, University of California, Santa Cruz

Santa Cruz, CA

- Designed an InGaAs/InP photodetector for telecommunications and semiconductor applications.

PROJECTS

BearWithMe.Tech | HackGT 13, AI & Web Development

- Won the MLH Best Domain from .tech award at HackGT 13, a hackathon with 1,300+ participants.
- Co-developed an AI-powered pediatric intake prototype integrating voice AI, computer vision, and real-time child, parent, and clinician interfaces using Next.js and Supabase.

RoboJackets RoboNav | Electrical Team

- Designed multilayer PCBs for power distribution, buck/boost regulation, and sensor breakout.
- Built wiring harnesses; validated motor drives with oscilloscopes and thermal imaging for URC field trials.

Athlosense | PCB Design, Sensors

- Developing a patent-pending, low-cost sports measurement device using custom PCBs and sensors to measure concussive and subconcussive forces in contact sports.

LEADERSHIP & EDITORIAL EXPERIENCE

Engineering Pivot | President and Director

- Directed a 1,200-member network across six continents, connecting 1,500+ students to mentored research and STEM publication opportunities reaching 300K+ readers.

Associate Editor

June 2025 – June 2026

Journal of Ethics in Scientific and Technological Innovations

Stony Brook, NY

- Served on the editorial team of a peer-reviewed scientific journal indexed by the Library of Congress.

SELECTED PUBLICATIONS

Scientific Reports: *Photoluminescence study of optical transitions in spinel zinc gallium oxide thin films*

Journal of Ethics in Scientific and Technological Innovations: *Ethical and Sustainability Challenges in the Global Semiconductor Supply Chain*

TECHNICAL SKILLS

Perception & ML: Computer vision, machine learning model development, large-scale transit data

Robotics Hardware: Multilayer PCB design, power distribution, buck/boost regulation, sensor breakout, motor drive circuits, wiring harnesses

Test & Measurement: Digital oscilloscopes, thermal imaging, function generators, digital multimeters

Semiconductors & Research: MBE, bandgap engineering, thin film characterization, photodetector design, experimental design, technical writing, peer review